

Agentic Specialist Capstone

Customer Escalation Triage

Student: Sai Krupa

A Hybrid Agentic Workflow for Faster, Safer Escalation Handling

Technology:

Zapier • Python • Gmail • Slack • Zapier Tables

Focus:

Quality • Speed • Safety • Traceability

Value Proposition

Automate repetitive escalation intake, classification, severity analysis, routing, approval-state management, and customer communication **while keeping high-risk customer-facing decisions under explicit human control.**

This capstone demonstrates a hybrid agentic workflow for customer escalation triage. The system combines three Zaps with a Python intelligence agent. Gmail provides the incoming escalation signal, Slack provides operational coordination and human approval, Zapier Tables provides persistent workflow state, and Python performs structured escalation analysis. The objective is not simply to automate customer communication. The objective is to automate the repetitive analysis and routing work while keeping human approval as a control point before customer communication.

BUSINESS PROBLEM & OPPORTUNITY

The Business Problem

Customer escalations require:

- Rapid identification and triage
- Consistent severity assessment
- Appropriate routing
- Careful customer communication
- Clear tracking of case status

Current Challenges

Manual effort

Repeated review and coordination

Inconsistent decisions

Severity and routing can vary

Communication risk

Customer responses require judgment

Limited visibility

Case state can span multiple tools

Opportunity

Automate repetitive work – without automating accountability.

Dimension	Baseline / Target	Acceptance Threshold
Quality	Manual/inconsistent	≥95% correct routing/severity on evaluation set
Speed	~15 min/escalation modeled baseline	<5 min automated processing target
Analyst effort	~25 hrs/week modeled baseline	≤10 hrs/week target
Governance	Manual approval	100% of high-risk external responses require approval
Auditability	Partial	Every escalation has structured state + decision history
Safety	Manual controls	Declined/unauthorized responses must not reach Gmail

Customer escalation handling contains a significant amount of repetitive coordination. Someone needs to identify the escalation, extract the important information, assess severity, determine routing, coordinate the response, and track the case. These activities create an opportunity for automation. However, customer communication is a higher-risk action, so the design intentionally keeps human accountability at that point.

GOALS & SUCCESS CRITERIA

What Does Success Look Like?

Dimension	Success Criteria
Quality	Structured and useful escalation analysis
Speed	Less repetitive manual triage and coordination
Safety	No customer response without required approval
Consistency	Repeatable severity and routing process
Traceability	Persistent case and decision state
Reliability	Representative tests + regression check

Key Principle

Measure more than automation.

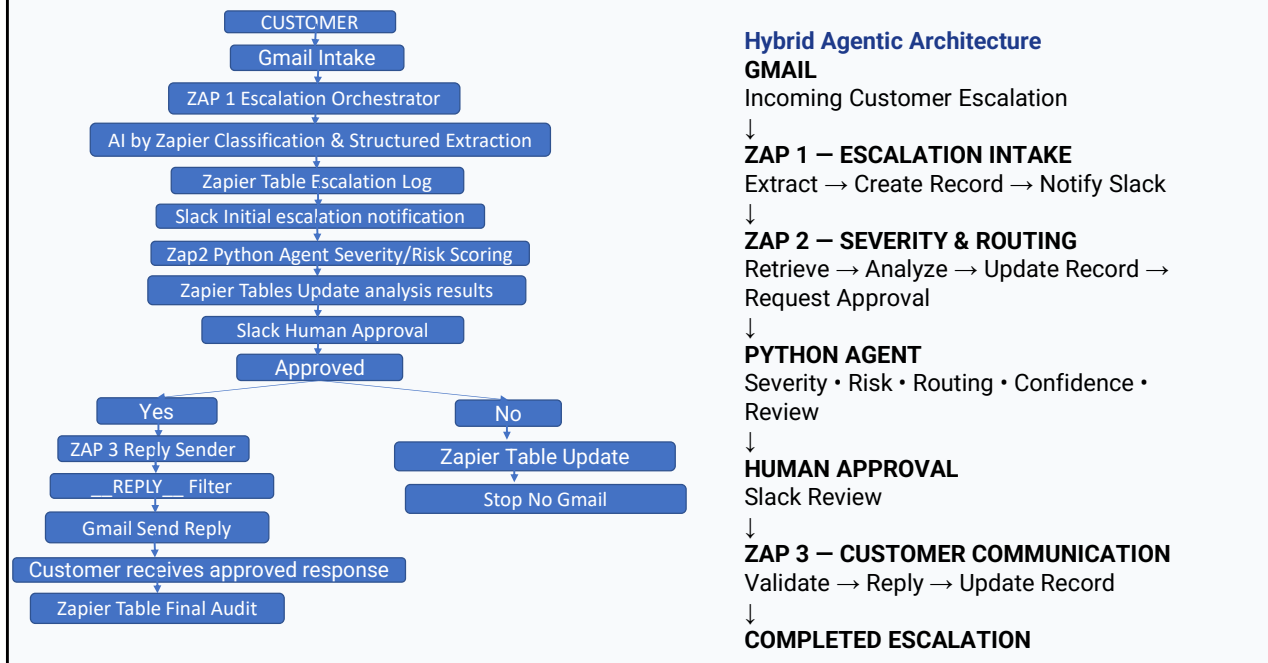
Quality + Speed + Safety + Traceability

The project defines success across multiple dimensions. Quality means that the escalation analysis is structured and useful. Speed means reducing repetitive triage and coordination work. Safety means preventing customer communication from bypassing approval.

Consistency comes from applying the same structured analysis process.

Traceability comes from maintaining a persistent escalation record. Reliability is demonstrated through representative testing and regression testing.

SOLUTION OVERVIEW & ARCHITECTURE



This is the overall architecture. Zap 1 handles intake and creates the persistent escalation record. Zap 2 retrieves that record and coordinates the analysis process. The Python agent provides the specialized escalation intelligence. The results are stored and presented for human review in Slack. Once approval is obtained, Zap 3 validates the approval and handles the customer reply. The workflow therefore separates intake, intelligence, human decision-making, and customer communication into controlled stages.

Zap 1: Escalation Intake

Zap 1 — Escalation Intake Orchestrator

Trigger

Gmail — New Message

Processing

1. Filter

Ignore messages labeled SENT

2. Extract

Convert incoming email into structured escalation information

3. Create Record

Store escalation in **Zapier Tables**

4. Notify

Post NEW CUSTOMER ESCALATION to Slack

5. Correlate

Store Slack timestamp with escalation record

Output

Persistent escalation record + Slack operational event

AS Capstone Zap 1 - Escalation Intake Orchestrator - v3.png



Zap 1 is the entry point into the system. It monitors Gmail for new messages and filters out messages that have already been sent. The email is converted into structured information. A persistent record is then created in Zapier Tables. The workflow posts a new escalation event to Slack so the operational team and downstream workflow can see the case. The Slack timestamp is also stored with the record, helping correlate the workflow across systems.

ZAP 2: SEVERITY & ROUTING

Zap 2 – Severity & Routing Worker

Trigger

Slack – New Escalation Event

Workflow

Identify

Extract escalation ID: ESC-\d+

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Retrieve

Find matching Zapier Tables record

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Analyze

Send escalation data to Python agent

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Receive

Severity • Risk • Routing • Confidence • Review

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Update

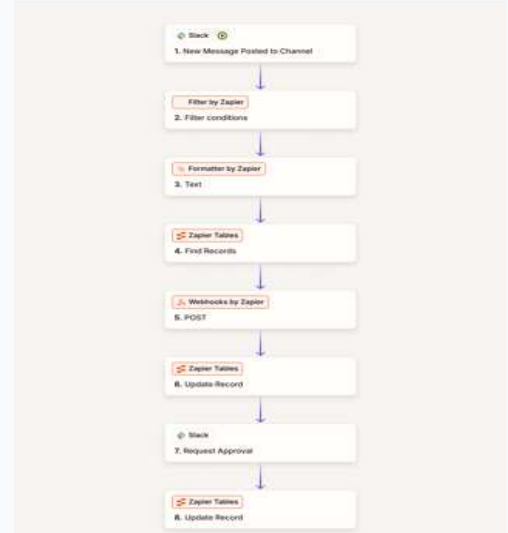
Write analysis back to Table

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Approve

Request human decision in Slack

AS Capstone Zap 2 - Severity and Routing Agent - v3.png



Zap 2 is the central analysis workflow. It starts when the new escalation event appears in Slack. The workflow extracts the escalation ID and uses that identifier to retrieve the correct persistent record. The relevant escalation information is then sent to the Python agent. The structured analysis is written back to the escalation record. Finally, the workflow requests a human approval decision in Slack.

Python Agent – Escalation Intelligence

What the Agent Produces

- **Severity / Priority**
How urgent is the escalation?
- **Risk**
What is the potential impact?
- **Routing**
Where should the case go?
- **Confidence**
How certain is the assessment?
- **Human Review**
Does the case require human attention?
- **Explanation**
Why was this assessment produced?

Representative Results

Scenario	Score	Priority
Production outage	95	P1
Billing dispute	90	P1
Security incident	80	P1
Enterprise bug	30	P3
General question	0	P4

[illegible]

Python provides the specialized intelligence component of the workflow. Instead of returning only a text response, the agent produces structured fields that can be consumed by the downstream automation. These include severity, risk, routing, confidence, human-review requirements, and an explanation. The representative test cases demonstrate differentiated outcomes. For example, a production outage received a score of 95 and P1 priority, while a general question received a score of 0 and P4 priority. This makes the agent useful as a decision-support component within the workflow.

HUMAN-IN-THE-LOOP + ZAP 3

Human Approval Controls Customer Communication

Automated

Analyze → Classify → Route → Prepare

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Human

Review → Approve / Decline

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Zap 3

Validate Approval → Send Reply → Update Record

Zap 3 – Controlled Customer Communication

Trigger: Slack __REPLY__

Validation:

Retrieve escalation + verify approval

Action:

Send Gmail reply using stored customer information

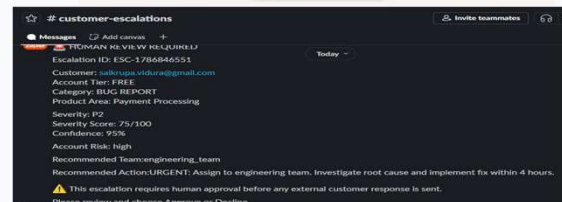
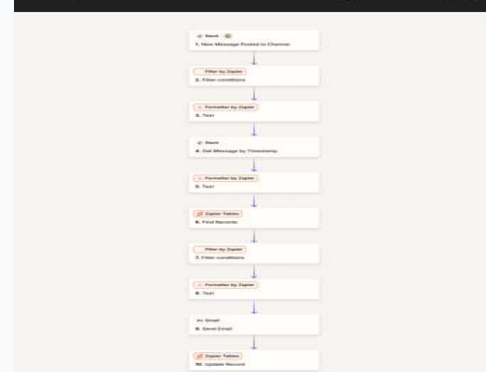
Completion:

Update Table with final status and timestamp

Key Control

No required approval = No customer reply

AS Capstone Zap 3 - Human Review & Reply Sender - v3.png



This is the key human-in-the-loop control. The automated system performs the repetitive analysis and prepares the workflow. A human reviews the result and decides whether the customer communication should proceed. Zap 3 is responsible for the actual customer response. It retrieves the escalation record and validates the approval state before sending. This means the system can automate communication execution without giving the agent unrestricted authority to communicate with customers.

EVALUATION & REGRESSION TESTING

Evaluation Evidence

Agent Evaluation

- Billing dispute → 90 / P1
- Production outage → 95 / P1
- Security incident → 80 / P1
- Enterprise bug → 30 / P3
- General question → 0 / P4

End-to-End Workflow

Approved Case

- ESC-1786846551
- Approved
- Customer response sent

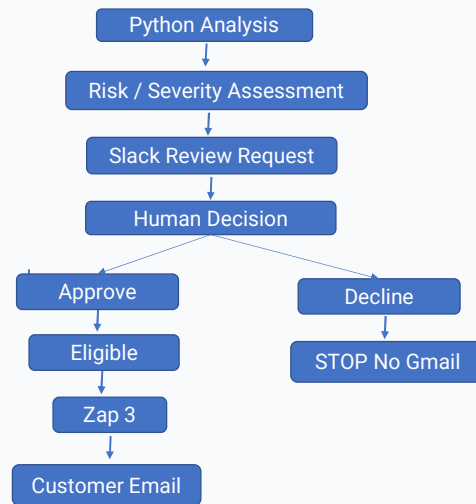
Declined Case

- ESC-1786847509
- Declined
- Customer response blocked

Regression

- Original subject preservation → PASS

Approval Flow



Evaluation was performed at both the agent and workflow levels. The agent was tested against multiple representative escalation scenarios with different priorities. We also tested the actual approval paths. ESC-1786846551 demonstrated the approved path where the customer response was sent. ESC-1786847509 demonstrated the declined path where the customer response was blocked. We also ran a regression test for original subject preservation, which passed. These tests provide evidence that the workflow is not simply executing — it is enforcing the intended decision paths.

SAFETY, GOVERNANCE & AUDITABILITY

Governance by Design

Input Controls

- Filter already-sent messages
- Structured extraction

State Controls

- Unique escalation ID
- Persistent Table record
- Correlated Slack events

Decision Controls

- Structured agent output
- Human approval
- Approval validation

Communication Controls

- Approval required before send
- Stored customer information
- Completion tracking

Auditability

Analysis results
Approval state
Final status
Timestamps

AS Capstone - Customer Escalation Log

Automate your data

Final Status	Reviewer Decision	Communication Status	Response Approval Status	Response Sent To
Completed	Approved	Sent	Approved	08/15/26
Completed	Approved	Sent	Approved	08/16/26
Declined	Declined	NOT STARTED	Declined	

Escalation ID	Customer Email	Original Subject	Action Taken	Issue Category	Issue Summary	Customer Impact	Urgency Signals	Product Area	Escalation Signal	Severity Score	Confidence Score	Account Health Level	Recommended Action	Human Review Requested	Reviewer Decision	Response Approved Status	Response Sent Time	Create Date	Slack Thread ID	Gmail Message ID
ESC-178684055	mlknq@all.com	URGENT - Test payment processing outage	FREE	BUG REPORT	Production payment environment is unavailable, preventing multiple customers from completing payments.	Multiple customers are unable to complete payment due to a production payment environment outage.	Subject line marked URGENT, production environment down, multiple customers affected, explicit request to investigate urgently.	Payments	TRUE	P2	75	95 high	URGENT: Assign to engineering team, investigate root cause and implement fix within 4 hours.	TRUE	Completed	Approved	8/16/26 12:05	8/16/26 12:05	178684055	14008546
ESC-178684700	mlknq@all.com	URGENT - Declined Escalation	FREE	BUG REPORT	A production payment issue has been reported and requires investigation. This is needed as a controlled and forced test of the declined customer escalation path.	Potential disruption to payment processing in a production environment.	Production environment affected, payment issue requiring investigation; escalation review requested.	Payments	TRUE	P2	65	90 high	URGENT: Assign to engineering team, investigate root cause and implement fix within 4 hours.	TRUE	Declined	NOT STARTED	8/16/26 12:05	8/16/26 12:05	178684700	14008566

Governance is built directly into the workflow. The system uses input controls to prevent duplicate processing. State controls allow each escalation to be tracked using a persistent identifier. Decision controls ensure that agent output does not automatically become customer communication. Communication controls validate approval before the Gmail action. Finally, the workflow stores analysis, approval, completion status, and timestamps to provide traceability.

BUSINESS VALUE & MEASUREMENT

Business Value

Reduce Manual Effort

Automate:

- Email extraction
- Escalation analysis
- Routing preparation
- Workflow coordination

Improve Speed

Immediate intake → Structured analysis

→ Faster review

Improve Consistency

Standardized analysis + repeatable workflow

Reduce Risk

Human approval + approval validation

Measure

Time / Escalation

Manual Touches

Approval Time

Rework

Classification Accuracy

Metric	Baseline	Target	Improvement
Processing time / escalation	~15 min	<5 min	~67% faster
Analyst effort	~25 hrs/week	~10 hrs/week	~60% reduction
Classification	Manual	Automated	Reduced repetitive work
Routing	Manual	Rule-based recommendation	More consistent
Audit	Partial	Structured table state	Improved traceability
Customer response	Manual	Approved automated workflow	Controlled execution

The value of this workflow comes from reducing repetitive operational work while maintaining control. Automated intake and analysis can reduce manual triage effort. Structured processing can also make the workflow faster and more consistent. The approval gate helps reduce communication risk. For production measurement, the key metrics should include time per escalation, manual touches, approval time, rework, and classification accuracy. These measurements allow the organization to determine whether the workflow is producing measurable operational improvement.

SCALE, ADOPTION & LAUNCH READINESS

From Capstone → Operational Workflow

1. VALIDATE

Controlled test cases
Quality + safety review

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2. PILOT

Limited operational group
Monitor results

↓

3. EXPAND

Increase volume
Refine routing and evaluation

↓

4. OPERATE

Continuous regression testing
Monitoring + governance

Launch Readiness

✓ 3-Zap workflow

✓ Python agent

✓ Persistent state

✓ Human approval gate

✓ Evaluation + regression testing

✓ Auditability

The recommended path to operational adoption is incremental. First, validate the workflow using controlled scenarios. Then move to a limited pilot with close human monitoring. Once quality and safety are demonstrated, expand the volume and routing coverage. In production, the workflow should continue to be monitored and regression-tested. The capstone implementation already demonstrates the major technical components, human approval control, evaluation approach, and auditability needed for that progression.

CONCLUSION

Key Takeaways

01

Automate repetitive escalation work

02

Use Python where specialized analysis adds value

03

Keep humans accountable for customer-facing decisions

04

Persist state so decisions are traceable

05

Measure quality, speed, safety, and business impact

Final Message

The strongest agentic workflows are not fully autonomous.

They are deliberately autonomous where automation is safe — and deliberately human where judgment matters.

This capstone demonstrates a practical hybrid agentic workflow rather than automation for automation's sake. Zapier provides orchestration, Python provides specialized escalation intelligence, Slack provides human coordination and approval, and Zapier Tables provides persistent state. The system automates repetitive work while preserving human accountability for customer-facing decisions. The key lesson is that effective agentic systems should be designed around measurable value, controlled autonomy, and traceability. That is the foundation for taking this workflow from a capstone demonstration toward an operational solution.